Liquidity Index Framework for an RPN Calculator

This document summarizes the proposed treatment of volume and liquidity in a mathematically consistent framework similar to logarithmic price returns. **Problem**
Price behaves naturally in logarithmic return space:

r_i = ln(P_i / P_(i-1))

These returns can be added, averaged, and accumulated. Volume does not have this property because it is scale-dependent. Adding the volume of Microsoft to the volume of a small mining stock has little meaning.

Dollar Volume (DV)
Instead of using raw volume, define:

DV = Price × Volume

This measures the amount of capital traded and is more comparable across stocks.

Average Dollar Volume
Use a rolling average, for example:

AvgDV60 = 60-day average(DV)
AvgDV100 = 100-day average(DV)

A 60-day window corresponds roughly to one quarter and is recommended as the primary measure.

Liquidity Deviation (LDV)
Define the liquidity deviation in logarithmic form:

LDV = ln(DV / AvgDV60)

Interpretation:
LDV = 0 : normal liquidity
LDV > 0 : above-normal participation
LDV < 0 : below-normal participation

Relationship to Relative Volume
The same concept can be written as:

Relative Dollar Volume = DV / AvgDV60
LDV = ln(Relative Dollar Volume)

This creates a dimensionless quantity that can be compared across stocks.

Liquidity Index Reconstruction
Just as price returns can be accumulated:

P_n = P_0 × exp(sum(r_i))

a liquidity index can be reconstructed:

$$L_n = L_0 \times \exp(\mathrm{sum}(LDV_i))$$

where L_0 is an arbitrary normalization constant, for example:

$$L_0 = 100$$

Interpretation of the Liquidity Index
A liquidity index above 100 indicates persistent above-average participation. A liquidity index below 100 indicates persistent below-average participation.

This is analogous to a normalized performance index rather than an actual traded value.

Suggested Use in the RPN Calculator
Treat price-return space and liquidity-participation space as parallel mathematical domains:

Price Return: $r_i = \ln(P_i / P_{(i-1)})$

Liquidity Participation: $LDV_i = \ln(DV_i / AvgDV60)$

Both quantities are logarithmic, additive, dimensionless, and compatible with the same mathematical operators.